Errata to Christodoulou (1993), found by a Lean formalization

D. Christodoulou, *Bounded variation solutions of the spherically symmetric Einstein-scalar field equations*, Comm. Pure Appl. Math. **46** (1993), 1131–1220.

Scope. This document records false statements or proof steps, substantive missing arguments, and misprints affecting mathematical content encountered in formalizing Sections 1 and 3–6. The numbering ERR-01–ERR-35 is stable. ERR-11 is a clarification of an explicitly stated normalization, not a defect; ERR-12 is a comparison with a later paper; ERR-34 records observations in Section 8, outside the formalization's scope. Routine typographical slips are not enumerated here. All page numbers are journal pages.

Theorem 3.1 (uniqueness and extension), Theorem 5.1, and the existence parts of Theorems 6.1 and 6.2 are proved in Lean, sealed, and independently audited. Their release declarations are `ChristodoulouBV.theorem3_1_unique`, `ChristodoulouBV.theorem3_1_extend`, `ChristodoulouBV.theorem5_1`, and `ChristodoulouBV.theorem6_1_6_2_exists`. **No defect below changes a main theorem as formalized.** The precise readings and the deferred finite-area conclusions are stated in the appendix.

The public evidence levels are:

- **Lean:** a sorry-free declaration in this repository proves the corrected form or refutes the printed one; the declaration is named in the entry.
- **reviewed:** an independent written review confirms the observation, without Lean evidence for that observation.
- **noted:** observed during the formalization, without independent written review of the observation.

A Lean proof of a correction does not by itself refute the printed statement. Entries distinguish these kinds of evidence. An **Affects** field identifies the argument needing repair, rather than an outstanding obligation in the released main theorems. Analytic area measures used in proofs should be distinguished from the paper's multiplicity image areas; see the appendix.

False statements and proof steps

ERR-01 — Lemma 6.2: the integrability conclusion is false

Location: p. 1189, Lemma 6.2 and the end of its proof.

The hypotheses give no uniform bound on the integrals of the nonnegative functions F_n. Thus Fatou's lemma gives an inequality in $[0,\infty]$, not $F \in L^1(0,1)$. There are data satisfying the hypotheses with every F_n integrable but $\int^- F = \infty$. The corrected conclusion is

$$\int^- F \le \liminf_{n\to\infty} \int^- F_n.$$

Here $\int^-$ is the lower Lebesgue integral; no measurability of F is asserted. Finiteness in applications must be obtained from the uniform smooth estimates.

Status: Lean, correction: `ChristodoulouBV.lemma6_2_lintegral`; counterexample to the printed integrability claim: `ChristodoulouBV.lemma6_2_L1_conclusion_false`.

Affects: the area-limit argument in the proofs of Theorems 6.1/6.2.

ERR-02 — Lemma 6.2: the lower-semicontinuity step fails

Location: p. 1189, proof of Lemma 6.2.

The essential bound on a rectangle controls almost every section, not every section outside the specified countable set. It therefore does not justify the claimed dominated-convergence argument at each such point. A counterexample has $y_n \in S$, $y_n \to y_0 \notin S$, $F(y_n) = 0$, and $F(y_0) = 1$, so lower semicontinuity fails. The example's F is measurable: it refutes the proof step, not measurability itself. Measurability from the printed hypotheses alone is neither proved nor refuted here. Use the lower-integral conclusion of ERR-01 and establish measurability separately for concrete applications.

Status: Lean, counterexample: `ChristodoulouBV.lemma6_2_lsc_step_false`; corrected conclusion: `ChristodoulouBV.lemma6_2_lintegral`.

Affects: the area-limit argument in the proofs of Theorems 6.1/6.2.

ERR-03 — Continuity of area from above requires finiteness

Location: pp. 1156–1157, opening area discussion in Section 4.

Nested domains with empty intersection need not have areas tending to zero if all their areas are infinite. For example, take the identity map on the tails $\{(u, v) : 0 < u < 1,\ v > n\}$. Add the premise that the first domain has finite area. For the Section 6 applications, finiteness must precede the shrinking-tail argument.

Status: Lean for the finite analytic-area correction: `ChristodoulouBV.analytic_area_continuity`. The counterexample to the unrestricted claim is **reviewed**, without a separate Lean refutation.

Affects: area tails in the proofs of Theorems 6.1/6.2; the multiplicity image-area conclusions remain deferred.

ERR-04 — The strict approximation bound fails when the variation is zero

Location: p. 1184, smooth approximation of the data for Theorem 6.2.

The printed $D_n < \min\{2D, \varepsilon_*\}$ is impossible for constant data, where $D = 0$. The approximation can instead satisfy $D_n \leq D$ and $D_n < \varepsilon_*$. Consequently $D_n < 2D$ when $D > 0$, while $D_n = 0$ when $D = 0$.

Status: Lean: `ChristodoulouBV.smooth_data_below_epsilon` (arbitrary point values) and `ChristodoulouBV.smooth_data_below_epsilon_of_right_continuous`.

Affects: the constant-data case in the proof of Theorem 6.2.

Substantive missing arguments

ERR-05 — Local existence at the axis needs a construction

Location: p. 1180, proof of Theorem 6.1; p. 1179, final continuation step in the proof of Theorem 5.1.

Local C^1 existence is invoked without a proof or reference. The boundary $r = 0$ is singular in the displayed equations, so this requires an axis-specific argument. The formalization supplies a quantitative Picard construction using r and $r\phi$, reconstructs the fields, and propagates the mass constraint. It also proves a lifetime uniform in a compact edge's C^1 bounds.

Status: Lean: `ChristodoulouBV.C1Solution.exists_local_c1_solution` and `ChristodoulouBV.C1Solution.exists_axis_triangle_solution_prescribed_compact`.

Affects: the proofs of Theorems 5.1 and 6.1.

ERR-06 — The comparison estimates depend on BV representatives

Location: pp. 1149–1155, the comparison norms and (3.13)–(3.21).

The BV definition imposes no representative convention. Equal initial r, ϕ give equal initial derivative fields only almost everywhere. Changing λ on one incoming null line can preserve the BV equations and r, ϕ, m while invalidating the printed pointwise comparison estimate (3.13). The suprema used there are therefore not representative-independent. Unqualified pointwise coincidence of all fields, including the conformal factor, has the same issue.

The formal proof uses a common conull set of initial coordinates, finite truncations, almost-everywhere transport identities, and recovery of r, ϕ, m from continuous primitives.

Status: Lean for the corrected uniqueness proof: `ChristodoulouBV.Uniqueness.ComparisonData.goodV_norms_zero`, `ChristodoulouBV.Uniqueness.ComparisonData.pointwise_of_goodV_norms_zero`, and `ChristodoulouBV.theorem3_1_unique`. The representative counterexample is **reviewed**, without a separate Lean refutation.

Affects: the proof and interpretation of Theorem 3.1; its formalized conclusion concerns r, ϕ, m.

ERR-07 — The uniform incoming derivative bound is not proved

Location: p. 1174, definition of $D'(u_0)$; its uses in (5.3), (5.10), and (5.12), pp. 1174–1177.

Individual C^1 solutions give bounds only on finite cutoffs. Finiteness of $D'(u_0)$ uniformly up to the terminal parameter requires an additional estimate; a direct bootstrap can be circular through its assumed lower bound for $1-\mu$. The formal proof first bounds the flux without dividing by $1-\mu$, then obtains lower expansion and mass-gap bounds and applies Gronwall to the derivative.

Status: Lean: `ChristodoulouBV.Uniqueness.incoming_weighted_bound_before_star`.

Affects: the proof of Theorem 5.1.

ERR-08 — Bounded derivatives do not supply a continuous terminal corner trace

Location: p. 1173 and p. 1177, after (5.12).

The asserted extension as a C^1 solution includes the closure and the terminal axis corner. The displayed derivative bounds alone do not prove continuity of the first derivatives there. The formal proof constructs off-axis terminal traces by parameter-dependent integral equations and crosses the corner using uniform local axis existence and C^1 uniqueness.

Status: Lean: `ChristodoulouBV.Uniqueness.exists_uniform_alphaPrime_bound_before_star`, `ChristodoulouBV.Uniqueness.exists_offAxisTerminalC1Packet`, and the extension in ERR-09.

Affects: the proof of Theorem 5.1.

ERR-09 — Strip continuation and gluing need additional justification

Location: pp. 1177–1179, continuation beyond v_*.

The strip argument invokes local continuation, a common strip width, corner compatibility, and gluing without supplying the needed theorems. A weighted bound on one derivative does not control all first derivatives. The formal proof uses an alternative construction: a lower cover with a full C^1 terminal seam, a uniform-lifetime axis triangle, identification by uniqueness, and gluing across the vertical seam. The paper's strip estimates are available but are not used in this construction.

Status: Lean: `ChristodoulouBV.Uniqueness.exists_routeB_extension`, yielding `ChristodoulouBV.theorem5_1`.

Affects: the proof of Theorem 5.1.

ERR-10 — The coefficient in (5.21) needs a sharper mass estimate

Location: p. 1179, mass increment estimate and (5.21).

The preceding coarse mass estimate yields a coefficient $8/9$ after multiplication by $2/r$, rather than the displayed $4/9$. The latter follows from the sharper estimate $\Delta m \le M^2(r^2-r_*^2)/9$, together with $m(u,v_*)\ge 0$. This sharper estimate is not printed.

Status: Lean: `ChristodoulouBV.Uniqueness.strip_mass_increment_sharp` and `ChristodoulouBV.Uniqueness.strip_mass_ratio_increment_le`.

Affects: the paper's strip proof, which the formal proof of Theorem 5.1 replaces as described in ERR-09.

ERR-11 — Scalar normalization in Section 4 (clarification, not a defect)

Location: pp. 1157 and 1159, the scalar-shift normalization and the estimate leading to (4.2a).

The paper explicitly uses scalar-shift invariance on p. 1157 to set $\phi(0,0)=0$ before deriving the amplitude estimate on p. 1159. The bound $|\alpha(0,v)|\le D$ uses this declared normalization. To apply the estimate to uncentered data, first center the scalar field and then shift the constructed solution back; the variation is unchanged. The entry is kept, under its stable number, because later uses of the Section 4 estimates must perform this centering.

Status: Lean for the paper's stated reduction: `ChristodoulouBV.s4_initial_alpha_abs_le` (with its explicit centering premise) and `ChristodoulouBV.scalar_shift`. This is not a counterexample and

not a missing premise in the printed argument.

Affects: reuse of the Section 4 estimates with uncentered data; no correction to the paper is needed.

ERR-12 — Theorem 6.2 does not state the quantitative conclusion cited later

Location: p. 1183, Theorem 6.2; compare Luk–Oh, Theorem 5.12, (5-29)–(5-31), and Remark 5.13, p. 1633.

The printed theorem asserts global BV existence and finite image areas. It does not explicitly state the bounds $1/3 \le \lambda \le 1/2$, $1/3 \le -\nu \le 2/3$, $2/3 \le 1-\mu \le 1$, the three-term linear variation bounds with one universal constant, or right-continuous representatives that Luk–Oh attribute to it. Extracting this stronger quantitative statement requires additional bookkeeping, including scalar-shift invariance and BV quotient estimates. This observation concerns the difference between the two stated conclusions; it is not a counterexample to either existence assertion.

Status: noted for this comparison of statements.

Affects: use of Theorem 6.2 as a citation for the stronger quantitative statement; no change to the formalized main theorems.

Reference: J. Luk and S.-J. Oh, Anal. PDE **8** (2015), 1603–1674, Theorem 5.12 and Remark 5.13.

ERR-13 — The area-limit applications require premises to be checked

Location: pp. 1190–1192, (6.4a)–(6.6), applications of Lemmas 6.1 and 6.2.

The argument does not establish the required differentiability of the limiting section, derivative convergence, one countable exceptional set valid for every section coordinate, integrability of the approximating functionals, or the interchange of truncation suprema and lower limits. The formal proof supplies absolutely continuous and almost-everywhere versions, regularizes the pointwise primitives, proves measurability for the actual functionals, and exhausts the cutoffs using uniform bounds. Coordinate factors are addressed in ERR-28.

Status: Lean: `ChristodoulouBV.area_lsc_regularization`, `ChristodoulouBV.common_section_measurability`, `ChristodoulouBV.area_chart_section_lintegral_le_liminf`, and `ChristodoulouBV.measure_global_le_of_areaOpenCutoff_bound`.

Affects: the limiting area estimates used to verify the BV conditions in Theorem 6.2; this does not assert the deferred image-area conclusions.

ERR-14 — The moving wedges are not nested

Location: p. 1182, area tails immediately before (6.1).

The rectangles $(0,u)\times(u,1)$ are not nested as u increases. Continuity from above therefore cannot be applied to them directly. Contain them in the nested tails of the full triangle on which $v > u$, and first prove that the full triangle has finite analytic area, as required by ERR-03.

Status: Lean: `ChristodoulouBV.s6_area_tail_geometry` and `ChristodoulouBV.s6_endpoint_tails`.

Affects: the terminal-tail argument in the proof of Theorem 6.1 and the smooth-solution construction used for Theorem 6.2.

ERR-15 — Representative changes must preserve the pointwise equations

Location: pp. 1196–1197, one-sided replacements of ν and β.

The existence of compatible traces on a common collection of lines, and the preservation of transport equations, mass and scalar-radius primitives, signs, and pointwise variation conditions, need proof. The formal construction chooses the incoming representatives simultaneously, proves stability of all primitive equations, and verifies the axis conditions for the same fields. It retains the prescribed outgoing data at every positive initial coordinate.

Status: Lean: `ChristodoulouBV.representative_stability`, `ChristodoulouBV.incoming_right_fields_item4`, and `ChristodoulouBV.representatives_incoming_corner`.

Affects: the BV witness in Theorem 6.2.

ERR-16 — Other missing arguments in Section 3

Location: pp. 1147–1155, Theorem 3.1 and its proof, including (3.1a,b) and the comparison estimates.

The following arguments are used without full justification: transferring the signs and mass-gap inequalities to BV solutions; taking axis limits in the logarithmic identities (3.1a,b); affine rescaling of BV solutions; finiteness of the mass comparison norms before absorption; regular Goursat uniqueness away from the axis; and gluing extensions with compatible BV representatives.

Status: Lean. Corresponding declarations include `ChristodoulouBV.IsBVSolution.mass_ratio_bounds`, `ChristodoulouBV.IsBVSolution.eq3_1a`, `ChristodoulouBV.IsBVSolution.eq3_1b`, `ChristodoulouBV.Uniqueness.rescaled_isBVSolution`, `ChristodoulouBV.Uniqueness.ComparisonData.Gamma_uniform_bound`, `ChristodoulouBV.Uniqueness.ComparisonData.GammaStar_uniform_bound`, `ChristodoulouBV.Uniqueness.regular_rectangle_unique`, and `ChristodoulouBV.Uniqueness.glued_isBVSolution`. The completed rescaling and continuation arguments are used by `ChristodoulouBV.theorem3_1_unique` and `ChristodoulouBV.theorem3_1_extend`.

Affects: the uniqueness and extension proofs of Theorem 3.1.

ERR-17 — Other missing arguments in Sections 1, 4, and 5

Location: pp. 1137–1138, (1.14)–(1.18); pp. 1162–1164, (4.7)–(4.11); pp. 1172–1177, Theorem 5.1 and (5.2)–(5.12).

Passing between the full and reduced systems requires nonvanishing of $1 - \mu$. The formalization takes the reduced system as its solution equations and proves the needed signs separately. The commutation of mixed derivatives used in Sections 4 and 5 needs regularity arguments. Treating a family as one solution requires compatibility, the C^1-to-BV implication, and agreement of initial scalar traces. Finally, terminal suprema must first be taken on finite cutoffs, before traces or uniform bounds are available.

Status: Lean for these supplied sign, regularity, compatibility, and cutoff arguments: `ChristodoulouBV.Uniqueness.nu_neg_of_c1_domain`, `ChristodoulouBV.Uniqueness.alphaPrime_deriv_u`, `ChristodoulouBV.Uniqueness.family_fields_eq_on_common_development`, `ChristodoulouBV.IsC1Solution.isBVSolution_development`, and `ChristodoulouBV.Uniqueness.weightedSup_self_bound`. The choice of reduced equations is an encoding convention, not a claim here of a separate formal equivalence theorem for the full system.

Affects: the proofs of Theorems 5.1 and 6.1.

ERR-18 — Other missing arguments in Section 6

Location: pp. 1180–1181, bootstrap; pp. 1184–1187, approximation and uniform bounds; pp. 1192–1193, BV transport identities (6.7)–(6.12).

The bootstrap needs a short initial interval and a first-exit argument for supremum-defined quantities. Smooth approximation must preserve arbitrary point values, including values at jumps. Uniform amplitude bounds need a datum offset: variation alone cannot bound constant data. Finally, the passage to BV transport and ratio estimates requires Stieltjes product and jump calculus. The formal proof supplies these arguments, keeping the amplitude offset out of the universal variation constants.

Status: Lean: `ChristodoulouBV.bootstrap_start`, `ChristodoulouBV.s6_exists_short_cone_sourceZ`, `ChristodoulouBV.exists_smooth_approx_arbitrary_values`, `ChristodoulouBV.uniform_smooth_bounds_for_sequence`, `ChristodoulouBV.bv_stieltjes_transport`, and `ChristodoulouBV.bv_ratio_bridge`. The bootstrap continuation is completed in `ChristodoulouBV.theorem6_1_6_2_exists`.

Affects: the proofs of Theorems 6.1/6.2.

Misprints affecting mathematical content

ERR-19 — The sign in (3.12)

Location: p. 1152, (3.12).

The incoming coefficient $\mu_i\nu_i/((1-\mu_i)r_i)$ is nonpositive. The printed upper bound on its integral is therefore vacuous; the comparison estimate needs the upper bound on the negative of that integral:

$$\sup_v \left[-\int_0^v \frac{\mu_i\nu_i}{(1-\mu_i)r_i}(u,v)\,du \right] \le c\varepsilon.$$

Status: Lean: `ChristodoulouBV.Uniqueness.ComparisonData.incomingCoeff_integral_small`.

Affects: the proof of Theorem 3.1.

ERR-20 — The incoming mass equation: denominator and integrating-factor reading

Location: p. 1154, incoming mass-difference equation and its integral formula.

The denominator multiplying ζ_2^2 in the differential equation must be ν_2, rather than λ_2. With $q = \zeta_2^2/(\nu_2 r_1) \le 0$, the terminal-value equation $\partial_u \Delta m = -q\Delta m + F/2$, $\Delta m(v,v) = 0$, has solution

$$\Delta m(u,v) = -\frac{1}{2}\int_u^v \exp\left(\int_u^{u'} q(u'',v)\,du'' \right) F(u',v)\,du'.$$

The exponential is at most one. The typography of the inner limits on pp. 1153–1154 is ambiguous: reading the incoming limits as u to u' requires changing the printed minus sign; reading them in the opposite order gives the same correct factor without a sign change. The denominator error is independent of this ambiguity. The equation reference is corrected in ERR-21.

Status: Lean for the corrected equation and terminal formula: `ChristodoulouBV.Uniqueness.ComparisonData.mass_in_difference_linear` and `ChristodoulouBV.Uniqueness.ComparisonData.mass_in_terminal_formula`. The ambiguity in the printed limits is **reviewed** and remains unresolved.

Affects: the incoming mass comparison in the proof of Theorem 3.1.

ERR-21 — Crossed equation references in Section 3

Location: pp. 1147, 1148, 1153, and 1154.

The outgoing logarithmic identity on p. 1147 should cite (1.14b), (1.21b); the incoming identity on p. 1148 should cite (1.14a), (1.21a). The outgoing mass-difference equation on p. 1153 should cite (1.15b); the incoming one on p. 1154 should cite (1.15a).

Status: reviewed. This is an observation about the printed references.

Affects: citation accuracy in the proof of Theorem 3.1.

ERR-22 — The axis relation has the wrong sign

Location: p. 1161, axis traces in Section 4.

The relation is $\nu|_\Gamma = -\lambda|_\Gamma$. The printed equality without the minus sign is incompatible with $r|_\Gamma = 0$ and with the subsequent relation $\beta|_\Gamma = -\alpha|_\Gamma$.

Status: Lean: `ChristodoulouBV.axis_traces`.

Affects: the Section 4 estimates used in the proofs of Theorems 6.1/6.2.

ERR-23 — Signs in the incoming logarithmic correction

Location: ppp. 1164–1165, the derivative of f^* and (4.11b)–(4.13b); p. 1169, (4.25a) and (4.26).

Writing J for the (u,v) Jacobian, the corrected derivative is

$$\partial_v f^* = \frac{\lambda\zeta^2/\nu - \nu\theta^2/\lambda}{(1-\mu)r^2} + J(\phi, \alpha/\lambda).$$

Both terms have the opposite sign to those printed. Consequently the quantity inside the v derivative in (4.11b) uses $+f^*$. The incoming residual in (4.13a), (4.13b), the Jacobian algebra preceding (4.25a), and (4.25a) is $\partial_u \log|\nu| - \mu\nu/((1-\mu)r) - f^*$. The derivation of (4.26) uses this corrected residual; its displayed unsigned area bound remains unchanged, as do the subsequent estimates. The correctly signed outgoing residual in (4.12b) is unaffected.

Status: **Lean**: `ChristodoulouBV.sourceLogCorrectionStar_vertical_deriv`, `ChristodoulouBV.eq4_11b_corrected`, `ChristodoulouBV.eq4_13b_corrected`, `ChristodoulouBV.eq4_25a`, and `ChristodoulouBV.eq4_26`.

Affects: intermediate Section 4 identities used in the proofs of Theorems 6.1/6.2.

ERR-24 — Missing minus signs in (5.4), (5.6), and (5.9)

Location: p. 1174, (5.4), (5.6); p. 1176, (5.9).

The middle lines require $-\int_{u_0}^{u} r^{-1-\delta}\partial_u r\, du'$. Without the minus sign they are nonpositive, since $\partial_u r = \nu < 0$, and cannot bound absolute values. The final bounds and coefficients are correct.

Status: **Lean** for the bounds using the correct signed integrals: `ChristodoulouBV.Uniqueness.integralI₁_le_weightedSup`, `ChristodoulouBV.Uniqueness.integralI₂_le_weightedSup`, and `ChristodoulouBV.Uniqueness.integralI₃_le_weightedSup`.

Affects: the proof of Theorem 5.1.

ERR-25 — The supremum before (5.10) needs a truncated domain

Location: p. 1176, immediately before (5.10).

Take the supremum over $\mathcal{D}(u_0, u_1, v_*)$, first with finite terminal cutoffs $t < v_*$. The printed full domain $\mathcal{D}(u_0, v_*)$ gives a supremum that may be infinite before absorption has proved a uniform bound.

Status: **Lean**: `ChristodoulouBV.Uniqueness.weightedSup_self_bound`.

Affects: the proof of Theorem 5.1.

ERR-26 — The strip domain and the closing strict inequality

Location: p. 1177, (5.13); p. 1178, definition of u_2; p. 1179, closing strip estimate.

In both (5.13) and the definition of u_2, the domain must be $\mathcal{D}(u_0, u_1, v_* + \eta)$. The printed domain ending at v_* does not include the new strip. In the closing estimate on p. 1179 the relation should be strict, $\sup_{U_\eta(u_2)} \mu < (\varepsilon + \varepsilon_1)/2$, rather than equality, as required by the preceding strict bound.

Status: **reviewed** for these corrections to the strip argument.

Affects: the paper's strip proof, replaced in the formal proof of Theorem 5.1 as described in ERR-09.

ERR-27 — The exponent in the lower bound for the incoming expansion

Location: p. 1181, bound for $\inf \nu$ in the bootstrap argument.

The exponent must have the positive sign: $\inf \nu \geq -\frac{1}{2}\exp(cD^2)$. The negative sign printed there would force $-\nu < 1/2$ uniformly for $D > 0$, contrary to its corner limit $1/2$.

Status: **Lean** for the corrected bootstrap bound: `ChristodoulouBV.s6_improve_null`; its proof obtains $-\nu \leq \frac{1}{2}\exp(CD^2)$ with positive C before deriving the stated uniform margin.

Affects: the proof of Theorem 6.1.

ERR-28 — Missing coordinate factors in the area substitutions

Location: pp. 1190–1191, substitutions for (6.4a)–(6.4d).

The transformed section density must contain the signed derivative v_x or u_x in both density terms. The transverse integration also contributes the factor $a-\delta$. The absolute Jacobians are $(a-\delta)^2(1-$

$y)$ and $(a-\delta)^2 y$ in the respective charts. Without these factors, the section functional is not the claimed area density.

Status: **Lean**: `ChristodoulouBV.area_section_primitive_eq_physical`, `ChristodoulouBV.areaUnitMap_abs_det`, and `ChristodoulouBV.area_cutoff_chart_lintegral`.

Affects: the limiting area estimates in the proof of Theorem 6.2.

ERR-29 — The shifted-edge term in (6.13a) is a residual variation

Location: p. 1194, (6.13a).

The right side cannot use the variation of the original datum α_0. Replacing it by the variation of $\alpha(a,\cdot)$ is still insufficient: $\lambda(a,\cdot)$ is generally not constant on a shifted edge. Retain instead the variation of the edge primitive of $d\alpha - \phi\, d\lambda$, as for the logarithmic residual in (6.14a). This edge residual has vanishing variation on the shrinking open interval, so (6.13b) and (6.15) follow.

Status: **Lean** for the corrected estimate and consequence: `ChristodoulouBV.eq6_13a` and `ChristodoulouBV.eq6_13b`. No Lean counterexample among full Einstein-scalar solutions is asserted here.

Affects: the axis variation conditions in the proof of Theorem 6.2.

ERR-30 — The reference before (6.14a)

Location: p. 1194, immediately before (6.14a).

The cited equation should be (6.8a), rather than (6.8b); the argument is the outgoing logarithmic residual estimate.

Status: **reviewed** for the printed cross-reference.

Affects: citation accuracy in the proof of Theorem 6.2.

ERR-31 — The zero conclusion is missing in (6.15)

Location: p. 1194, (6.15).

Both displayed limits equal zero. Merely equating them, as printed, omits the conclusion required for the axis variation condition and established by the preceding residual estimates and smallness absorption.

Status: **Lean**: `ChristodoulouBV.eq6_15` proves both zero limits.

Affects: the axis variation conditions in the proof of Theorem 6.2.

ERR-32 — The residual term is missing in (6.17b)

Location: p. 1195, (6.17b).

The integrand must be $|d\log|\nu| + g^*\, du|$, as in (6.17a), rather than the variation of $\log|\nu|$ alone. It is this residual that the preceding transport argument controls and the proof of (6.18) consumes.

Status: **Lean**: `ChristodoulouBV.eq6_17b`, with the corrected residual.

Affects: the incoming axis variation conditions in the proof of Theorem 6.2.

ERR-33 — The final BV condition reference

Location: p. 1197, after the limit of $\alpha + \beta$ at the axis.

The limit verifies condition 6 of the BV definition on p. 1147, rather than condition 4. Condition 4 concerns shrinking variation near the axis.

Status: **reviewed** for the printed cross-reference.

Affects: citation accuracy in the proof of Theorem 6.2.

ERR-34 — Three sign observations in Section 8

Location: p. 1218, trapped-region bound for $r\lambda$; p. 1219, its oscillation bound and the analogous bound for $r\nu$.

In the first and third bounds, delete the leading minus sign in front of the factor $1-r/r_0$. As printed, the lower bound is positive although the quantity being bounded is negative. In the second bound, insert a minus in front of the right-hand factor $r\lambda$: the left side is nonnegative, whereas the printed right side is negative.

Status: noted; these observations have no independent written review and no Lean evidence.

Affects: Section 8, which is outside this formalization's scope.

ERR-35 — The incoming area comparison must retain terminal-edge jumps

Location: p. 1195, (6.17a), and its use in (6.17b)–(6.18), pp. 1195–1196.

The incoming Stieltjes transport involves a half-open strip including its terminal characteristic. A jump there can give the analytic area measure an atom that the printed open triangle misses. Use the half-open triangle $\{a-\varepsilon < u < v \le a\}$ instead. For sufficiently small $\varepsilon > 0$ these sets lie in the global domain, have finite analytic area, and decrease to the empty set. Their areas therefore tend to zero, giving the required residual limits.

Status: Lean for the corrected estimate: `ChristodoulouBV.eq6_17a`; continuity on these triangles: `ChristodoulouBV.bv_analytic_area_incoming_half_open_triangle_tendsto`. This correction does not claim a Lean counterexample to (6.17a) among full Einstein-scalar solutions.

Affects: the incoming axis variation conditions in the proof of Theorem 6.2.

Appendix: Readings and encoding choices (not defects)

- **Theorem 3.1.** Uniqueness is pointwise for the geometric and scalar quantities r, ϕ, m. The derivative fields and conformal factor require representative qualifications; see ERR-06. The extension conclusion uses the same interpretation.
- **Theorems 6.1/6.2.** The release proves global C^1 and BV existence with the exact initial traces and one common positive smallness threshold. Their printed finite multiplicity image-area conclusions are deferred. The analytic area measures used internally to prove the BV conditions do not constitute those deferred conclusions.
- **Lemmas 6.1/6.2.** The section functional is the variation of the pointwise integration-by-parts primitive $G(x) = f(x)g(x) - f(0)g(0) - \int_0^x (gf' + h)$, and L^∞ bounds are essential bounds. Lemma 6.2 is used only as $\int^- F \le \liminf_n \int^- F_n$, with no measurability claim for F. The failed lower-semicontinuity step remains recorded in ERR-02. Measurability for concrete section functionals is proved separately.
- **Area in the BV proof.** A countably additive analytic area measure is obtained by integrating section variation measures. Pointwise closed-section variation and this measure can differ at jumps. Domination by the pointwise functional is used on open cutoffs; closed cutoffs are contained in larger open cutoffs before taking limits. These are proof choices, separate from the paper's image-area conclusions.
- **Initial data in Theorem 6.2.** There is no right-continuity premise. The initial trace of α is prescribed at every $v > 0$, including arbitrary values at jumps. Choosing incoming representatives does not replace these outgoing data.
- **Domain and equations.** The global domain is $\{(u, v) : 0 \le u < v\}$, including the initial edge, with gauge $r(0, v) = v/2$. The solution predicates encode the reduced equations by interval primitive identities. The C^1 regularity requirement is on the closure of the development domain.
- **Scope exclusions.** Sections 2, 7, and 8 are not formalized. ERR-34 is retained as an explicitly noted observation. Errors in the formalization's own preliminary arguments or administrative records are not paper errata.